

CSA 0378-DATA STRUCTURES

8. Single linked list- insertion at Beginning, at End, and at ANY POINT

```
main.c
1 #include<stdio.h>
2 #include<stdlib.h>
3
4 struct node
5 {
6     int data;
7     struct node *next;
8 };
9
10 struct node *head = NULL;
11
12 // Insert at Beginning
13 void insertBeginning(int val)
14 {
15     struct node *newnode = (struct node*)malloc(sizeof(struct node));
16     newnode->data = val;
17     newnode->next = head;
18     head = newnode;
19     printf("%d inserted at beginning\n", val);
20 }
21
22 // Insert at End
23 void insertEnd(int val)
24 {
25     struct node *newnode = (struct node*)malloc(sizeof(struct node));
26     newnode->data = val;
27     newnode->next = NULL;
28
29     if(head == NULL)
30     {
31         head = newnode;
```

```
Output
Clear
Linked List: 4 -> NULL
--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 1
Enter value: 5
5 inserted at beginning
--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 2
Enter value: 7
7 inserted at end
--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice:
```

```
main.c
34 }
35
36 struct node *temp = head;
37 while(temp->next != NULL)
38     temp = temp->next;
39
40 temp->next = newnode;
41 printf("%d inserted at end\n", val);
42 }
43
44 // Insert at Any Position
45 void insertPosition(int val, int pos)
46 {
47     struct node *newnode = (struct node*)malloc(sizeof(struct node));
48     newnode->data = val;
49     newnode->next = NULL;
50
51     if(pos == 1)
52     {
53         newnode->next = head;
54         head = newnode;
55         printf("%d inserted at position %d\n", val, pos);
56         return;
57     }
58
59     struct node *temp = head;
60     int i;
61     for(i=1; i<pos-1; i++)
62     {
63         if(temp == NULL)
64             return;
```

```
Output
Clear
Linked List: 4 -> NULL
--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 1
Enter value: 5
5 inserted at beginning
--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 2
Enter value: 7
7 inserted at end
--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice:
```

```
main.c
68 }
69     temp = temp->next;
70 }
71
72 newnode->next = temp->next;
73 temp->next = newnode;
74 printf("%d inserted at position %d\n", val, pos);
75 }
76
77 // Display
78 void display()
79 {
80     struct node *temp = head;
81
82     if(temp == NULL)
83     {
84         printf("List is empty!\n");
85         return;
86     }
87
88     printf("Linked List: ");
89     while(temp != NULL)
90     {
91         printf("%d -> ", temp->data);
92         temp = temp->next;
93     }
94     printf("NULL\n");
95 }
96
97 void main()
98 {
```

```
Output
Clear
Linked List: 4 -> NULL
--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 1
Enter value: 5
5 inserted at beginning
--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 2
Enter value: 7
7 inserted at end
--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice:
```

main.c

Share

Run

Output

Clear

```
97 void main()
98 {
99     int choice, val, pos;
100
101     while(1)
102     {
103         printf("\n--- Single Linked List Menu ---\n");
104         printf("1. Insert at Beginning\n");
105         printf("2. Insert at End\n");
106         printf("3. Insert at Any Position\n");
107         printf("4. Display\n");
108         printf("5. Exit\n");
109         printf("Enter choice: ");
110         scanf("%d", &choice);
111
112         switch(choice)
113         {
114             case 1:
115                 printf("Enter value: ");
116                 scanf("%d", &val);
117                 insertBeginning(val);
118                 break;
119
120             case 2:
121                 printf("Enter value: ");
122                 scanf("%d", &val);
123                 insertEnd(val);
124                 break;
125
126             case 3:
127                 printf("Enter value: ");
```

2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 4
Linked List: 4 -> NULL

--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 1
Enter value: 5
5 inserted at beginning

--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 2
Enter value:
7
7 inserted at end

--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End

main.c

Share

Run

Output

Clear

```
115         printf("Enter value: ");
116         scanf("%d", &val);
117         insertBeginning(val);
118         break;
119
120     case 2:
121         printf("Enter value: ");
122         scanf("%d", &val);
123         insertEnd(val);
124         break;
125
126     case 3:
127         printf("Enter value: ");
128         scanf("%d", &val);
129         printf("Enter position: ");
130         scanf("%d", &pos);
131         insertPosition(val, pos);
132         break;
133
134     case 4:
135         display();
136         break;
137
138     case 5:
139         exit(0);
140
141     default:
142         printf("Invalid choice!\n");
143 }
144 }
145 }
```

2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 4
Linked List: 4 -> NULL

--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 1
Enter value: 5
5 inserted at beginning

--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Insert at Any Position
4. Display
5. Exit
Enter choice: 2
Enter value:
7
7 inserted at end

--- Single Linked List Menu ---
1. Insert at Beginning
2. Insert at End